

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Both needed for one mark X = (gill) filaments Y = (gill) arch/bar	1			1		1
		(ii)		{Filter/sieve/remove/catch} {food/particles/solids NOT molecules} /protection from debris/preventing damage to the gill filaments (1)		1		1		
		(iii)		A. normally water separates {filaments/lamellae/plates/gills}/Gills dry out (when out of water) (1) B. {Filaments/lamellae/plates/gills} {stick together/collapse} (1) C. Reduced Surface Area for {diffusion/gas exchange} (1) D. Water needed for <u>diffusion</u> (of gases)/{oxygen/gases} need to be in solution to <u>diffuse</u> (1)	4			4		
	(b)			Accept reverse argument for any point A. Countercurrent more efficient (1) B. Maintains {diffusion/concentration} gradient/or description of/equilibrium not met (1) C. Across {entire/whole} {gill/filament/lamellae/plate} (1) D. {Higher/more} {saturation/oxygen absorbed} (1) Accept correct references to higher percentages in counter current/use of values e.g. 80% compared to 50%	4			4		
	(c)	(i)		C = parallel flow and A = countercurrent flow (1)		1		1		
		(ii)		Correct insertion of arrows on A <u>only</u> (1) NOT ECF	1			1		

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	(d)	(i)		Positive (1)			1	1	1	
		(ii)		Range of m = 0.6 – 0.65 = 2 marks Accept correct fraction on answer line number of dp = neutral If incorrect award one mark for sight of dy/dx (dy is smaller than dx) (1)		2		2	2	
		(iii)		Accept range 96-104 a.u. (1) (value from (ii) x 160) + 0 ECF		1		1	1	
		(iv)		Any two (x1) from A. wider the { lumen/blood vessel} faster {velocity/flow rate/speed} (1) Reject faster velocity the wider the lumen B. so blood can be returned to the heart as quickly as possible (1) C. without {losing velocity/slowing down} due to decreased blood pressure/because no contractions to increase pressure (1)			2	2		
				Question 4 total	10	5	3	18	4	1